Power Analysis for Longitudinal Clinical Trials with Run-In and Common Close Observations

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Abstract

Background. Longitudinal clinical trials often incorporate pre-randomization (run-in) and post-treatment (common close) observations to improve estimator efficiency. Closed-form power and sample-size formulas for these augmented designs under linear mixed-effects models with random slopes have remained scattered across the literature and partially specified, making applied sample-size calculation cumbersome.

Methods. We derive closed-form power and sample-size formulas for three nested cases: (i) two run-in observations, (ii) an arbitrary number J_0 of run-in observations, and (iii) J_0 run-in plus J_2 common close observations. All derivations proceed via the Woodbury matrix identity applied to the GLS estimator under change-score parameterization. The formulas are expressed in terms of the residual variance σ^2, the random-slope variance σ_b^2, and the observation spacing t, enabling direct computation of required sample sizes.

Results. In a numerical example using parameters from Alzheimer's disease neuroimaging trials, for which the signal-to-noise ratio $\sigma_b/\sigma \approx 1.7$ is favorable, two run-in observations reduce the required sample size by 62% (from 385 to 145 per group), and adding two common close observations raises the reduction to 82%. The magnitude of the gain is governed by σ_b/σ: across a representative range of slope variability the reduction from two run-in observations spans roughly 14% at low ratios to 76% at high ratios. The closed-form expressions allow this sensitivity analysis with respect to σ_b/σ without re-running simulation.

Conclusions. The Woodbury-based formulas give closed-form expressions for the variance of the treatment-effect estimator under random-slopes longitudinal-trial designs with run-in and common close observations. They serve as a foundation for the AR(1) extension (Paper 2), the optimal-allocation problem (Paper 3), and the GLS-versus-ANCOVA-versus-averaged estimator comparison (Paper 4) in this compendium.

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1 Introduction

[1] **Beyond its traditional roles, the run-in period has a distinct methodological value in rate-of-change trials: estimating individual baseline trajectories to raise power.**

- Conventional run-in uses include identifying compliant participants and washing out prior treatments.
- When the outcome is a rate of change, run-in observations estimate each participant's baseline trajectory.
- Incorporating those trajectories into the analysis model substantially increases statistical power.

The run-in period has served multiple purposes in clinical trial design, from identifying compliant participants to washing out prior treatments[1]*. A dis-*

tinct and methodologically important use has emerged in trials where the primary outcome is a rate of change: estimating individual baseline trajectories that, when incorporated into the analysis model, substantially increase statistical power.

[2] **Frost and colleagues established the mixed-model foundation: run-in observations help most when slope variability is large relative to measurement error.**

- The key insight is that pre-randomization observations inform each participant's underlying trajectory.
- This reduces the residual variance of the treatment-effect estimator, and the gain grows with the slope-to-noise ratio.
- In Alzheimer's neuroimaging trials, where that ratio is favorable, they reported sample-size reductions of 30 to 45 percent.

[2] established the theoretical foundation for this approach within the linear mixed effects framework. Their key insight was that when between-subject variability in rates of change is large relative to measurement error, pre-randomization observations provide valuable information about each participant's underlying trajectory, reducing the residual variance of the treatment effect estimator. Working with neuroimaging outcomes in Alzheimer's disease trials, where the ratio of slope variability to measurement error is particularly favorable, they demonstrated sample size reductions of 30–45%.

[3] **Wang and colleagues extended the framework with two-period models that jointly model run-in and post-randomization data rather than treating run-in as a covariate.**

- Joint modelling of both periods yields additional power gains of up to 15 percent.
- It permits unequal randomization ratios without power loss.
- Run-in observations contribute placebo-equivalent information for all participants.

[3] extended this framework by proposing two-period linear mixed effects models that jointly model run-in and post-randomization data rather than using run-in observations merely as covariates. Their approach yields additional power gains of up to 15% and enables unequal randomization ratios without power loss, since run-in observations effectively contribute placebo-equivalent information for all participants.

[4] **A parallel literature gives closed-form variance and sample-size formulae for the random-coefficient model without a run-in phase, which form the baseline this paper extends.**

- Lu and colleagues derived the cLDA estimator variance, its link to ANCOVA, and the conditions under which the two coincide.
- Hu and colleagues generalized these to random-coefficient models

with monotone missing data and variable follow-up, illustrated with ADNI data.
- These results cover only the $J_0 = 0$ case and serve as the baseline extended here.

A parallel literature has developed closed-form variance and sample size formulae for the random coefficient regression model without a run-in phase.[4] *derived the variance of the constrained longitudinal data analysis estimator and its relation to analysis of covariance, and*[5] *established efficiency conditions under which the two approaches coincide.*[6] *generalized these results to random coefficient regression models with monotone missing data and variable follow-up, providing variance, power, and sample size expressions illustrated with data from the Alzheimer's Disease Neuroimaging Initiative. These formulae apply to designs with $J_0 = 0$ run-in observations and an arbitrary post-randomization visit schedule, and serve as the baseline case that we extend here.*

[5] **The CRAN longpower package is the reference implementation of the baseline formulae, and our general result reduces to its routines at the appropriate boundaries.**

- longpower provides edland, diggle, and liu.liang routines for slope-comparison trials.
- The Stata package slopepower offers a complementary simulation-based route to the same class of calculations.
- Our formula reduces to the Ard-Edland slope variance at $J_0 = J_2 = 0$, and to the Diggle compound-symmetry form when also $\sigma_b^2 = 0$.
- The generalization thus embeds the existing longpower formulae as the baseline against which run-in and common-close gains are measured.

The CRAN reference implementation of the $J_0 = 0$ formulae is the `longpower` package[7]*, which provides the routines `edland.linear.power()`, `diggle.linear.power()`, and `liu.liang.linear.power()` for slope-comparison trials with random intercepts and slopes*[8–10]*. A complementary implementation is the Stata package `slopepower`*[11]*, which obtains power for slope-comparison trials by simulating from a fitted mixed model rather than by evaluating a closed form; it addresses the same design question but does not cover run-in or common close phases, and its simulation route does not expose the variance as an explicit function of the design counts. We show in Section 3.3 that the closed-form result derived below reduces to the Ard-Edland slope-variance formula implemented in `edland.linear.power()` at the boundary $J_0 = J_2 = 0$, and to the Diggle compound-symmetry formula when additionally $\sigma_b^2 = 0$. The generalization developed here therefore embeds the existing `longpower` formulae as the baseline case against which the efficiency gains from run-in and common close observations are measured.*

[6] **No prior closed-form power formula handles an arbitrary number of run-in observations together with a separate**

common-close phase, the gap this paper fills.

- The common-close phase observes all participants off-treatment after the randomized period.
- It supplies extra information about individual trajectories at no cost to the treatment comparison.
- Such post-treatment periods appear in delayed-start and randomized-withdrawal designs, but their GLS efficiency contribution has not been formally characterized.

Despite these advances, no general closed-form power formula has been derived that simultaneously accounts for an arbitrary number of run-in observations and a separate set of common close observations collected after treatment cessation. The common close phase, in which all participants are observed off-treatment following the randomized period, provides additional information about individual trajectories at no cost to the treatment comparison. Post-treatment observation periods appear in delayed-start and randomized withdrawal designs aimed at distinguishing symptomatic from disease-modifying effects[12], but the contribution of the common close to statistical efficiency within the GLS framework has not been formally characterized. The delayed-start design[13] is the closest relative: it also observes participants after the initial randomized period, but it does so in order to test whether an early-treatment group retains an advantage over a late-start group, so the post-randomization phase carries its own treatment contrast and its own estimand. The common close considered here serves the different purpose of sharpening the estimate of each participant's underlying slope, with all participants off treatment and no contrast attached to the phase itself. A formal comparison of the two estimands, and of the assumptions each requires about post-cessation trajectories, is beyond the scope of this paper.

[7] **The paper presents three nested results, all derived from the GLS estimator via the Woodbury identity.**

- A closed-form variance for exactly two run-in observations (Section 3.2), then its generalization to arbitrary J_0 (Section 3.3).
- A further extension adding J_2 common-close observations (Section 3.4).
- All expressions are in terms of σ^2, σ_b^2, and spacing t, with numerical examples from Alzheimer's neuroimaging studies (Section 5).

We present three results. First, we derive a simple closed-form variance formula for the treatment effect estimator when exactly two run-in observations precede randomization (Section 3.2). Second, we generalize this formula to an arbitrary number J_0 of run-in observations (Section 3.3). Third, we further extend to include J_2 common close observations collected after treatment cessation (Section 3.4). All derivations proceed from the generalized least squares estimator under the Woodbury matrix identity, yielding expressions in terms of the residual variance σ^2, the random slope variance σ_b^2, and the observation spacing t. Section 5 provides numerical examples using parameters estimated from Alzheimer's disease neuroimaging studies.

2 Model

2.1 General formulation

[8] **The model accommodates a two-arm longitudinal trial whose observation schedule spans three distinct phases.**

- N participants are observed at equally spaced time points.
- The phases are run-in (pre-randomization), treatment (post-randomization), and an optional common close (post-treatment).
- Y_{ij} denotes the outcome for participant i at time point j.

Consider a two-arm longitudinal clinical trial in which N participants are observed at equally spaced time points across three phases: a run-in period (pre-randomization), a treatment period (post-randomization), and an optional common close period (post-treatment). Let Y_{ij} denote the outcome for participant i at time point j. The observation schedule is:

- **Run-in:** $j = -J_0, -J_0 + 1, \dots, -1$ (all participants untreated)
- **Randomization visit:** $j = 0$ (baseline)
- **Treatment period:** $j = 1, 2, \dots, J_1$ (participants randomized to treatment or placebo)
- **Common close:** $j = J_1+1, J_1+2, \dots, J_1+J_2$ (all participants off-treatment)

Throughout, the phase counts (J_0, J_1, J_2) correspond one-to-one to the arguments `r` (run-in), `k` (post-randomization), and `f` (common close) of the accompanying `runinpower` package functions used in the code chunks below.

We model the outcome as

$$Y_{ij} = \alpha + a_i + (\delta(1 - h_j) + \beta h_j + \gamma h_j g_i + b_i)t_j + u_{ij} + \varepsilon_{ij} \tag{1}$$

[9] **The outcome model combines a random intercept and random slope with a three-component within-subject variance structure following Frost and colleagues.**

- a_i and b_i are the random intercept and slope, while u_{ij} and ε_{ij} are the between-visit and between-scan residuals.
- Both residual terms are independent across visits, so under a single scan per visit they enter every formula only through their sum.
- We therefore define the total within-visit residual $w_{ij} = u_{ij} + \varepsilon_{ij}$ with variance σ^2.

where α is the overall intercept, $a_i \sim N(0, \sigma_a^2)$ is a random intercept, $b_i \sim N(0, \sigma_b^2)$ is a random slope, $u_{ij} \sim N(0, \sigma_u^2)$ are within-subject between-visit random effects independent across visits, and $\varepsilon_{ij} \sim N(0, \sigma_\varepsilon^2)$ are independent between-scan measurement errors. This three-component within-subject variance structure follows[2] (their Eq. 17). Because u_{ij} and ε_{ij} are both independent across visits, under a single scan per visit they enter every formula below only through their sum. We therefore write $w_{ij} = u_{ij} + \varepsilon_{ij}$ for the total *within-visit residual, with*

$$\sigma^2 \;\equiv\; \mathrm{Var}(w_{ij}) \;=\; \sigma_u^2 + \sigma_\varepsilon^2. \tag{2}$$

[10] **The two within-visit components are identifiable only with replicate scans, so single-scan designs depend solely on their sum.**

- Separating σ_u^2 from σ_ε^2 requires a visit carrying replicate scans.
- For the single-scan designs treated here, only the sum σ^2 enters the treatment-effect variance.
- This reduction simplifies all subsequent variance expressions.

The two within-visit components are separately identifiable only when a visit carries replicate scans; for the single-scan designs treated here only their sum σ^2 enters the variance of the treatment-effect estimator. The phase indicator is

$$h_j = \begin{cases} 0 & \text{if } j \le 0 \quad \text{(run-in)} \\ 1 & \text{if } j > 0 \quad \text{(post-randomization)} \end{cases}$$

and the treatment group indicator is

$$g_i = \begin{cases} 1 & i = 1, \dots, m \quad \text{(treatment)} \\ 0 & i = m+1, \dots, N \quad \text{(placebo)} \end{cases}$$

[11] **Three fixed-effect slope parameters describe the run-in, placebo post-randomization, and treatment trajectories under equally spaced observations.**

- δ is the run-in rate of change common to all participants.
- β is the placebo post-randomization rate of change, and γ is the treatment effect on that rate.
- Observations are equally spaced with interval t, so $t_j = jt$.

The fixed effects are: δ, the rate of change during the run-in period (common to all participants); β, the rate of change during the post-randomization period for the placebo group; and γ, the treatment effect on the rate of change. We assume equally spaced observations with interval t, so $t_j = jt$.

[12] **During the common close both arms revert to a shared off-treatment slope, modelled by deactivating the treatment indicator in that phase.**

- For $j > J_1$ all participants are off-treatment with slope $\beta + b_i$.
- This is implemented by setting $g_i = 0$ for every participant during the common close.
- Equivalently, the treatment indicator is replaced with $g_i \cdot I(1 \le j \le J_1)$ in the model.

For the common close phase ($j > J_1$), both treatment and placebo groups revert to a common off-treatment slope. We model this by setting $g_i = 0$ for all participants during the common close, so that the slope for both groups is $\beta + b_i$ in this phase. This is equivalent to replacing the treatment indicator with $g_i \cdot I(1 \leq j \leq J_1)$ in model (1).

2.2 Change-score parameterization

Following[2], we work with change scores relative to the randomization visit:

$$c_{ij} = Y_{ij} - Y_{i0} \tag{3}$$

[13] **Differencing relative to baseline removes the intercept and aligns the analysis with constrained longitudinal data analysis under compound symmetry.**

- The change score eliminates α and the random intercept a_i, leaving only two variance components.
- It is algebraically equivalent to cLDA, so the derived formulas apply to both analyses of run-in designs.
- By dropping σ_a^2 it avoids the anticonservative sample sizes that arise from ignoring intercept-slope correlation.

This eliminates the intercept α and the random intercept a_i. We note that the change-score approach is algebraically equivalent to constrained longitudinal data analysis (cLDA) under compound symmetry, where the baseline is included in the response vector with a common-mean constraint across treatment groups[14,15]. The variance formulas derived below therefore apply to both change-score and cLDA analyses of run-in designs. Because the change-score eliminates a_i, the resulting formulas depend on only two variance components (σ^2, σ_b^2) rather than the three ($\sigma^2, \sigma_a^2, \sigma_b^2$) required by the raw-outcome model.[16] show that ignoring the intercept-slope correlation in the three-component model can produce anticonservative sample sizes; the change-score approach avoids this issue by construction, at the cost of discarding information about σ_a^2.

The change-score formulation yields

$$c_{ij} = (\delta(1 - h_j) + \beta h_j + \gamma h_j g_i + b_i)t_j + (w_{ij} - w_{i0}) \tag{4}$$

for $j \neq 0$. The change-score vector for participant i has dimension $J = J_0 + J_1 + J_2$, encompassing all three phases.

2.3 Covariance structure

The marginal covariance of the change-score vector is

$$\Sigma = R + ZGZ' \tag{5}$$

where $G = \sigma_b^2$ (scalar, since we have a single random slope), $Z = t \cdot \mathbf{1}_J$ is the $J \times 1$ random effects design vector, and R is the $J \times J$ residual covariance of the differenced within-visit residuals $(w_{ij} - w_{i0})$.

The residual covariance has the structure

$$R_{jl} = \mathrm{Cov}(w_{ij} - w_{i0},\ w_{il} - w_{i0}) = \begin{cases} 2\sigma^2 & \text{if } j = l \\ \sigma^2 & \text{if } j \neq l \end{cases} \tag{6}$$

[14] **The shared baseline residual induces compound-symmetry correlation among change scores, so the residual covariance depends only on the summed within-visit variance.**

- The within-visit residuals are independent across visits with common variance σ^2.
- Differencing against the shared baseline w_{i0} produces positive correlation, giving $R = \sigma^2(\mathbf{I}_J + \mathbf{1}_J\mathbf{1}_J')$.
- Both within-visit components enter R identically, so only their sum is identified from single-scan designs.

since $w_{ij} = u_{ij} + \varepsilon_{ij}$ are independent across visits with common variance $\sigma^2 = \sigma_u^2 + \sigma_\varepsilon^2$, and the shared baseline residual w_{i0} induces positive correlation among all change scores. Thus $R = \sigma^2(\mathbf{I}_J + \mathbf{1}_J\mathbf{1}_J')$. Both within-visit components enter R identically because u_{i0} and ε_{i0} are differenced out of every change score in the same way; this is why only their sum σ^2 is identified from single-scan designs.

3 Variance of the Treatment Effect Estimator

3.1 GLS estimation and the Woodbury identity

The generalized least squares estimator of the fixed effects is

$$\hat{\beta} = (X'\Sigma^{-1}X)^{-1}X'\Sigma^{-1}\mathbf{c} \tag{7}$$

[15] **The estimand of interest is the treatment-effect variance, a single diagonal element of the inverted GLS information matrix.**

- The GLS estimator has covariance $(X'\Sigma^{-1}X)^{-1}$.
- X is the fixed-effects design matrix and $\mathbf{c}$ is the stacked change-score vector.
- The target $\mathrm{Var}(\hat{\gamma})$ is the diagonal entry corresponding to γ.

with $\mathrm{Var}(\hat{\beta}) = (X'\Sigma^{-1}X)^{-1}$, where X is the fixed effects design matrix and $\mathbf{c}$ is the stacked change-score vector. Our target is $\mathrm{Var}(\hat{\gamma})$, the variance of the treatment effect estimator, which is the diagonal element of $(X'\Sigma^{-1}X)^{-1}$ corresponding to γ.

The key computational tool is the Woodbury matrix identity. Since $\Sigma = R + ZGZ'$

with scalar $G = \sigma_b^2$:

$$\Sigma^{-1} = R^{-1} - R^{-1}Z\left(G^{-1} + Z'R^{-1}Z\right)^{-1}Z'R^{-1} \tag{8}$$

This avoids direct inversion of the $J \times J$ matrix Σ and enables closed-form results because R^{-1} has a known simple structure.

From $R = \sigma^2(\mathbf{I}_J + \mathbf{1}_J\mathbf{1}_J')$, the Sherman-Morrison formula gives

$$R^{-1} = \frac{1}{\sigma^2}\left(\mathbf{I}_J - \frac{1}{J+1}\mathbf{1}_J\mathbf{1}_J'\right) \tag{9}$$

With $Z = t \cdot \mathbf{1}_J$, we obtain

$$Z'R^{-1}Z = \frac{t^2}{\sigma^2}\left(J - \frac{J^2}{J+1}\right) = \frac{J\,t^2}{\sigma^2(J+1)} \tag{10}$$

and

$$\left(G^{-1} + Z'R^{-1}Z\right)^{-1} = \frac{\sigma^2\sigma_b^2(J+1)}{\sigma^2(J+1) + J\,t^2\sigma_b^2} \tag{11}$$

Let $a = \sigma^2$ and $b = \sigma_b^2$ for conciseness. Define

$$\eta_J = \frac{a(J+1) + J\,t^2 b}{a(J+1)} = 1 + \frac{J\,t^2 b}{a(J+1)} \tag{12}$$

so that $(G^{-1} + Z'R^{-1}Z)^{-1} = b/\eta_J$.

3.1.1 Per-participant information, compact form

Let X_i, Z_i, and R_i denote the per-participant fixed effects design, random effects design, and residual covariance, and define the scalar

$$\mathcal{J}_i = Z_i'R_i^{-1}Z_i + G^{-1} \tag{13}$$

By (8), the per-participant contribution to the GLS information matrix is

$$X_i'\Sigma_i^{-1}X_i = X_i'R_i^{-1}X_i - \frac{(X_i'R_i^{-1}Z_i)(Z_i'R_i^{-1}X_i)}{\mathcal{J}_i} \tag{14}$$

[16] **The additive per-participant information form has two properties exploited later: a scalar rank-one correction and accommodation of heterogeneous designs.**

- The total information is the sum of per-participant contributions, with covariance given by its inverse.
- Under a single random slope, $\mathcal{J}_i$ is a scalar, making the correction a rank-one adjustment of the OLS-style information.
- Because contributions add, varying J_0, J_2, or dropout patterns are handled by summing over strata.

and the total information is $X'\Sigma^{-1}X = \sum_i X_i'\Sigma_i^{-1}X_i$, with $\mathrm{Var}(\hat{\beta}) = (X'\Sigma^{-1}X)^{-1}$. Two features of this form are used later: (i) under a single random slope, $\mathcal{J}_i$ is a scalar so (14) is a rank-one correction of the OLS-style information $X_i'R_i^{-1}X_i$; (ii) because per-participant contributions add, heterogeneous designs (varying J_0, J_2, or dropout patterns) are accommodated by summing (14) over the relevant strata.

3.2 Case 1: Standard design, no run-in ($J_0 = 0,\ J_1 = 2,\ J_2 = 0$)

[17] **The no-run-in case reproduces established results and serves as a verification baseline for the framework.**

- It reproduces Frost and colleagues, Section 3.1.1.
- It is the $J_0 = 0$ baseline of the random-coefficient power formulae of Lu and Hu and colleagues.
- It thereby provides a verification checkpoint for the present derivation.

This case reproduces the result in[2], Section 3.1.1, and is the $J_0 = 0$ baseline of the random coefficient regression power formulae of[4] and[6]. It serves as verification for our framework.

[18] **Without a run-in phase the design reduces to two post-randomization change scores and a two-parameter fixed-effects model.**

- There are $J = 2$ change scores at post-randomization times t and $2t$.
- The absence of run-in removes the δ parameter from the model.
- The remaining fixed effects are the placebo slope β and the treatment effect γ.

With $J_0 = 0$ and $J_1 = 2$, there are $J = 2$ change scores $(c_{i,1}, c_{i,2})$ at post-randomization times $(t, 2t)$. Since there is no run-in, the model has two fixed effects: the placebo slope β and the treatment effect γ.

The design matrices are:

$$X_i^{(\mathrm{trt})} = t\begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix}, \qquad X_i^{(\mathrm{plc})} = t\begin{pmatrix} 1 & 0 \\ 2 & 0 \end{pmatrix}$$

The residual covariance is $R = a\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ with $R^{-1} = \frac{1}{3a}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$, and the random effects design is $Z = t(1,\, 2)'$.

Applying the Woodbury identity with $J = 2$:

$$Z'R^{-1}Z = \frac{t^2}{3a}(1,\, 2)\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}\begin{pmatrix} 1 \\ 2 \end{pmatrix} = \frac{t^2}{3a}(1,\, 2)(0,\, 3)' = \frac{6t^2}{3a} = \frac{2t^2}{a} \tag{15}$$

$$(G^{-1} + Z'R^{-1}Z)^{-1} = \frac{ab}{a + 2bt^2} \tag{16}$$

After summing over one treatment and one placebo participant, the information matrix is

$$X'\Sigma^{-1}X = \frac{2t^2}{a + 2bt^2} \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} \tag{17}$$

and its inverse is

$$\mathrm{Var}(\hat{\beta}) = \frac{a + 2bt^2}{2t^2} \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix} \tag{18}$$

The variance of the treatment effect estimator (per participant pair) is the $(2, 2)$ element:

$$\boxed{\mathrm{Var}(\hat{\gamma})\big|_{J_0=0, J_1=2} = \frac{a + 2bt^2}{t^2} = \frac{\sigma^2 + 2\sigma_b^2 t^2}{t^2}} \tag{19}$$

[19] **The recovered variance matches Frost and supplies the per-group baseline for all run-in comparisons.**

- The expression agrees with Frost and colleagues, Section 3.1.1.
- For m participants per group the variance scales as $(\sigma^2 + 2\sigma_b^2 t^2)/(mt^2)$.
- This quantity is the reference against which run-in designs are evaluated.

This agrees with Frost et al. (2008), Section 3.1.1. For m participants per group: $\mathrm{Var}(\hat{\gamma}) = (\sigma^2 + 2\sigma_b^2 t^2)/(mt^2)$. This serves as the baseline against which the run-in designs are compared.

3.3 Case 2: Two run-in observations ($J_0 = 2,\ J_1 = 1,\ J_2 = 0$)

[20] **Two run-in observations yield three change scores and a design matrix that distinguishes the run-in slope from the post-randomization slopes.**

- There are $J = 3$ change scores spanning the pre- and post-randomization times.
- The fixed-effects model now includes the run-in slope δ alongside β and γ.
- The design matrix separates the run-in contribution from the post-randomization contribution.

We now have $J = 3$ change scores $(c_{i,-2},\ c_{i,-1},\ c_{i,1})$ at times $(-2t,\ -t,\ t)$, the baseline visit at time 0 having been differenced away. The fixed effects design matrix separates the run-in slope δ from the post-randomization slopes:

$$X_i^{(\mathrm{trt})} = \begin{pmatrix} -2t & 0 & 0 \\ -t & 0 & 0 \\ 0 & t & t \end{pmatrix}, \qquad X_i^{(\mathrm{plc})} = \begin{pmatrix} -2t & 0 & 0 \\ -t & 0 & 0 \\ 0 & t & 0 \end{pmatrix}$$

[21] **Because the random slope acts uniformly across visits, the random-effects design vector collapses to the scaled time distances from baseline.**

- The random slope b_i applies to all time points alike.
- The random-effects design therefore reduces to $Z = t(-2, -1, 1)'$.
- The run-in change scores are differences of the pre-randomization outcomes from baseline.

Since the random slope applies uniformly to all time points, $Z = t(-2, -1, 1)'$ for the random effect b_i. We note that the change scores at times $-2t$ and $-t$ relative to baseline at 0 yield $c_{i,-2} = Y_{i,-2} - Y_{i,0}$ and $c_{i,-1} = Y_{i,-1} - Y_{i,0}$.

More carefully, under model (1):

$$
\begin{aligned}
c_{i,-2} &= -2(\delta + b_i)\,t + (w_{i,-2} - w_{i,0})\\
c_{i,-1} &= -(\delta + b_i)\,t + (w_{i,-1} - w_{i,0})\\
c_{i,1} &= (\beta + \gamma g_i + b_i)\,t + (w_{i,1} - w_{i,0})
\end{aligned}
\tag{20}
$$

Thus the design matrices for the fixed effects $(\delta, \beta, \gamma)'$ are:

$$
X_i^{(\mathrm{trt})} = t\begin{pmatrix} -2 & 0 & 0\\ -1 & 0 & 0\\ 0 & 1 & 1 \end{pmatrix}, \qquad X_i^{(\mathrm{plc})} = t\begin{pmatrix} -2 & 0 & 0\\ -1 & 0 & 0\\ 0 & 1 & 0 \end{pmatrix}
$$

and the random effects design is $Z_i = t(-2, -1, 1)'$.

The residual covariance is

$$
R = a\begin{pmatrix} 2 & 1 & 1\\ 1 & 2 & 1\\ 1 & 1 & 2 \end{pmatrix} = a(\mathbf{I}_3 + \mathbf{1}_3\mathbf{1}_3') \tag{21}
$$

with inverse

$$
R^{-1} = \frac{1}{a}\left(\mathbf{I}_3 - \frac{1}{4}\mathbf{1}_3\mathbf{1}_3'\right) = \frac{1}{4a}\begin{pmatrix} 3 & -1 & -1\\ -1 & 3 & -1\\ -1 & -1 & 3 \end{pmatrix} \tag{22}
$$

Now we compute the Woodbury components. With $Z = t(-2, -1, 1)'$:

$$
Z'R^{-1}Z = \frac{t^2}{4a}(3\cdot 4 + 3\cdot 1 + 3\cdot 1 - 2(-1)(-2)(-1) - 2(-1)(-2)(1) - 2(-1)(-1)(1)) \tag{23}
$$

Computing directly: $Z'R^{-1}Z = \frac{t^2}{4a}(-2, -1, 1)\begin{pmatrix} 3 & -1 & -1\\ -1 & 3 & -1\\ -1 & -1 & 3 \end{pmatrix}(-2, -1, 1)'$

$$R^{-1}Z = \frac{t}{4a}\begin{pmatrix} 3(-2)+(-1)(-1)+(-1)(1) \\ (-1)(-2)+3(-1)+(-1)(1) \\ (-1)(-2)+(-1)(-1)+3(1) \end{pmatrix} = \frac{t}{4a}\begin{pmatrix} -6 \\ -2 \\ 6 \end{pmatrix}$$

so $Z'R^{-1}Z = \frac{t^2}{4a}[(-2)(-6)+(-1)(-2)+(1)(6)] = \frac{t^2}{4a}(12+2+6) = \frac{20t^2}{4a} = \frac{5t^2}{a}$.

Alternatively, we note $Z = t \cdot \mathbf{d}$ where $\mathbf{d} = (-2,-1,1)'$, so $Z'R^{-1}Z = \frac{t^2}{a}(\mathbf{d}'\mathbf{d} - \frac{1}{4}(\mathbf{1}'\mathbf{d})^2) = \frac{t^2}{a}(6 - \frac{4}{4}) = \frac{5t^2}{a}$.

Thus

$$H \equiv (G^{-1} + Z'R^{-1}Z)^{-1} = \frac{ab}{a+5bt^2} \tag{24}$$

For the information matrix, we compute $X'R^{-1}X$ summed over treatment and placebo groups. For a single treatment participant:

$$X_i^{(\mathrm{trt})\prime} R^{-1} X_i^{(\mathrm{trt})} = \frac{t^2}{4a}\begin{pmatrix} -2 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} 3 & -1 & -1 \\ -1 & 3 & -1 \\ -1 & -1 & 3 \end{pmatrix}\begin{pmatrix} -2 & 0 & 0 \\ -1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}$$

Computing the product:

$$X_i^{(\mathrm{trt})\prime} R^{-1} X_i^{(\mathrm{trt})} = \frac{t^2}{4a}\begin{pmatrix} 11 & 3 & 3 \\ 3 & 3 & 3 \\ 3 & 3 & 3 \end{pmatrix} \tag{25}$$

For placebo (whose γ column is zero):

$$X_i^{(\mathrm{plc})\prime} R^{-1} X_i^{(\mathrm{plc})} = \frac{t^2}{4a}\begin{pmatrix} 11 & 3 & 0 \\ 3 & 3 & 0 \\ 0 & 0 & 0 \end{pmatrix} \tag{26}$$

Summing (one participant per group for the derivation, then multiplying by m):

$$X'R^{-1}X = \frac{t^2}{4a}\begin{pmatrix} 22 & 6 & 3 \\ 6 & 6 & 3 \\ 3 & 3 & 3 \end{pmatrix} \tag{27}$$

For the Woodbury correction, we need $X'R^{-1}Z$. For treatment:

$$X_i^{(\mathrm{trt})\prime} R^{-1} Z = \frac{t^2}{4a}\begin{pmatrix} -2 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} -6 \\ -2 \\ 6 \end{pmatrix} = \frac{t^2}{4a}\begin{pmatrix} 14 \\ 6 \\ 6 \end{pmatrix}$$

For placebo:

$$X_i^{(\mathrm{plc})\prime} R^{-1} Z = \frac{t^2}{4a} \begin{pmatrix} -2 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} -6 \\ -2 \\ 6 \end{pmatrix} = \frac{t^2}{4a} \begin{pmatrix} 14 \\ 6 \\ 0 \end{pmatrix}$$

The Woodbury correction requires the per-subject outer products (not the outer product of the sum):

$$\begin{aligned} W &= \sum_g X_i^{(g)\prime} R^{-1} Z \cdot H \cdot Z' R^{-1} X_i^{(g)} \\ &= \frac{bt^4}{16a(a+5bt^2)} \left(\begin{pmatrix} 14 \\ 6 \\ 6 \end{pmatrix} (14, 6, 6) + \begin{pmatrix} 14 \\ 6 \\ 0 \end{pmatrix} (14, 6, 0) \right) \\ &= \frac{bt^4}{16a(a+5bt^2)} \begin{pmatrix} 392 & 168 & 84 \\ 168 & 72 & 36 \\ 84 & 36 & 36 \end{pmatrix} \end{aligned} \tag{28}$$

Therefore

$$X' \Sigma^{-1} X = X' R^{-1} X - W \tag{29}$$

> [22] **The treatment-effect variance is the relevant diagonal element of the inverted information matrix, obtained numerically and checked against the closed form.**
>
> - $\mathrm{Var}(\hat{\gamma})$ is the $(3,3)$ element of $(X'\Sigma^{-1}X)^{-1}$.
> - It is evaluated numerically in the accompanying R code.
> - The numerical value is verified against the derived closed-form result.
>
> *To find $\mathrm{Var}(\hat{\gamma})$, we need the $(3,3)$ element of $(X'\Sigma^{-1}X)^{-1}$. We compute this numerically in the R code below and verify against the closed-form result.*

> [23] **Completing the cofactor and determinant algebra yields the closed-form treatment-effect variance for the two-run-in design.**
>
> - Define the auxiliary quantity $D = \sigma^2 + 5\sigma_b^2 t^2$.
> - The $(3,3)$ cofactor simplifies to $6t^4/(aD)$.
> - Dividing the cofactor by the determinant gives the boxed variance below.
>
> *Carrying through the algebra, define $D = a + 5bt^2 = \sigma^2 + 5\sigma_b^2 t^2$ and write $X'\Sigma^{-1}X = \frac{t^2}{16aD} A$. The matrix A has $(3,3)$ minor $1536\, aD$ and determinant $9216\, aD\,(a+2bt^2)$. Hence the $(3,3)$ cofactor of $X'\Sigma^{-1}X$ is $1536aD \cdot t^4/(16aD)^2 = 6t^4/(aD)$, and its determinant is $9216\, aD\,(a+2bt^2) \cdot t^6/(16aD)^3$. The ratio gives:*

$$\boxed{\mathrm{Var}(\hat{\gamma})\big|_{J_0=2,J_1=1} = \frac{8\sigma^2(\sigma^2+5\sigma_b^2t^2)}{3t^2(\sigma^2+2\sigma_b^2t^2)}} \tag{30}$$

[24] **The run-in variance differs structurally from the Frost baseline by involving the residual variance in both numerator and denominator.**

- In the Frost formula σ^2 appears only additively in the numerator.
- Here σ^2 enters both the numerator and the denominator.
- This reflects the extra information extracted from pre-randomization observations.

Unlike the Frost formula (19), the run-in variance depends on σ^2 both in the numerator and denominator, reflecting the additional information extracted from pre-randomization observations.

3.4 General formula: J_0 run-in observations $(J_2 = 0)$

[25] **The derivation now generalizes to an arbitrary number of run-in observations with no common close.**

- The design has arbitrary J_0 run-in and J_1 post-randomization observations.
- The common close is absent, so $J_2 = 0$.
- The total number of change scores is $J = J_0 + J_1$.

We now derive the variance for arbitrary J_0 run-in observations with J_1 post-randomization observations and no common close ($J_2 = 0$). The total number of change scores is $J = J_0 + J_1$.

The change scores are

$$\begin{aligned} c_{i,-s} &= -s(\delta + b_i)\,t + (w_{i,-s} - w_{i0}) && s = 1, \dots, J_0 \\ c_{i,l} &= l(\beta + \gamma g_i + b_i)\,t + (w_{i,l} - w_{i0}) && l = 1, \dots, J_1 \end{aligned} \tag{31}$$

[26] **The design matrix is built from run-in and post-randomization time vectors mapping to the three fixed-effect columns.**

- The columns correspond to the parameters δ, β, and γ.
- $\mathbf{d}_{J_0}$ collects the negative run-in times scaled by t.
- $\mathbf{d}_{J_1}$ collects the positive post-randomization times scaled by t.

The fixed effects design matrix has columns for (δ, β, γ). Let $\mathbf{d}_{J_0} = (-J_0, -J_0+1, \dots, -1)' \cdot t$ be the run-in time vector and $\mathbf{d}_{J_1} = (1, 2, \dots, J_1)' \cdot t$ be the post-randomization time vector. Then:

$$X_i^{(\mathrm{trt})} = \begin{pmatrix} \mathbf{d}_{J_0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{d}_{J_1} & \mathbf{d}_{J_1} \end{pmatrix}, \qquad X_i^{(\mathrm{plc})} = \begin{pmatrix} \mathbf{d}_{J_0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{d}_{J_1} & \mathbf{0} \end{pmatrix}$$

The random effects design vector is $Z_i = (\mathbf{d}'_{J_0}, \mathbf{d}'_{J_1})'$, i.e., the full vector of time distances from baseline.

The residual covariance remains $R = a(\mathbf{I}_J + \mathbf{1}_J\mathbf{1}'_J)$ with inverse given by (9).

3.4.1 Computing $X'\Sigma^{-1}X$

[27] **The Woodbury computation is organized around the sum-of-squares of the run-in and post-randomization time indices.**

- The derivation proceeds via the Woodbury identity.
- S_{J_0} is the sum of squared run-in indices in closed form.
- S_{J_1} is the corresponding sum over post-randomization indices.

We proceed through the Woodbury identity. Define $S_{J_0} = \sum_{s=1}^{J_0} s^2 = J_0(J_0+1)(2J_0+1)/6$ *and* $S_{J_1} = \sum_{l=1}^{J_1} l^2 = J_1(J_1+1)(2J_1+1)/6$.

The key intermediate quantities are:

$Z'R^{-1}Z$: With $Z = t \cdot \mathbf{d}$ where $\mathbf{d} = (-J_0, \dots, -1, 1, \dots, J_1)'$:

$$Z'R^{-1}Z = \frac{t^2}{a}\left[(S_{J_0} + S_{J_1}) - \frac{(\sum d_j)^2}{J+1}\right] \tag{32}$$

where $\sum d_j = -J_0(J_0+1)/2 + J_1(J_1+1)/2$.

$(G^{-1} + Z'R^{-1}Z)^{-1}$: With $Z'R^{-1}Z$ given by (32),

$$H = \left(G^{-1} + Z'R^{-1}Z\right)^{-1} = \frac{b}{1 + b\,Z'R^{-1}Z},$$

the general analogue of b/η_J in (12).

$X'R^{-1}X$: This 3×3 matrix has elements computed from the block structure of X. Letting $D_{J_0} = \sum_{s=1}^{J_0} s = J_0(J_0+1)/2$ and $D_{J_1} = \sum_{l=1}^{J_1} l = J_1(J_1+1)/2$, and using $R^{-1} = \frac{1}{a}(\mathbf{I}_J - \frac{1}{J+1}\mathbf{1}\mathbf{1}')$, the information matrix summed over one treatment and one placebo participant is:

$$X'R^{-1}X = \frac{t^2}{a}\begin{pmatrix} 2\left(S_{J_0} - \frac{D_{J_0}^2}{J+1}\right) & \frac{2D_{J_0}D_{J_1}}{J+1} & \frac{D_{J_0}D_{J_1}}{J+1} \\ \frac{2D_{J_0}D_{J_1}}{J+1} & 2\left(S_{J_1} - \frac{D_{J_1}^2}{J+1}\right) & S_{J_1} - \frac{D_{J_1}^2}{J+1} \\ \frac{D_{J_0}D_{J_1}}{J+1} & S_{J_1} - \frac{D_{J_1}^2}{J+1} & S_{J_1} - \frac{D_{J_1}^2}{J+1} \end{pmatrix} \tag{33}$$

[28] **The structure of the information matrix reflects which parameters both groups inform and the opposing signs of the two phase time-sums.**

- Entries informed by both groups carry a factor of 2.
- The γ-related entries lack this factor because the placebo γ column is zero.

- The positive (δ, β) cross-term arises from the opposite signs of the run-in and post-randomization time sums.

The factor of 2 appears on entries involving both groups, while γ-related entries lack this factor since the γ column is zero for the placebo group. The positive sign on the (δ, β) cross-term arises because the run-in and post-randomization time sums have opposite signs ($\sum d_{J_0} < 0$, $\sum d_{J_1} > 0$).

Woodbury correction: The correction is the sum of per-subject rank-one outer products:

$$W = \sum_{g \in \{\text{trt}, \text{plc}\}} X_i^{(g)\prime} R^{-1} Z \cdot H \cdot Z' R^{-1} X_i^{(g)}$$

which is generally rank 2 (not rank 1), so $X'\Sigma^{-1}X = X'R^{-1}X - W$.

3.4.2 Numerical implementation

[29] **For general run-in and post-randomization counts the variance has no compact closed form and must be evaluated as a complex ratio.**

- No simple closed-form expression exists for arbitrary J_0 and J_1.
- The variance depends on the sum-of-squares and sum quantities together with the dimension J.
- The Woodbury correction interacts with the information matrix to produce algebraically complex expressions.

The variance $\mathrm{Var}(\hat{\gamma})$ for general J_0 and J_1 does not admit a simple closed-form expression. The formula depends on σ^2 and σ_b^2 in a ratio involving S_{J_0}, S_{J_1}, D_{J_0}, D_{J_1}, and J, and the interaction between the Woodbury correction and the information matrix produces algebraically complex expressions.

[30] **A matrix-based R implementation evaluates the exact variance for any configuration, with the closed-form cases as checkpoints.**

- The computation uses the matrix Woodbury identity in R code.
- It returns the exact variance for any (J_0, J_1, J_2) configuration.
- The $J_0 = 0$ and $J_0 = 2$ closed forms serve as verification checkpoints.

We therefore implement the computation via the matrix-based Woodbury identity in R code, which evaluates the exact variance for any configuration (J_0, J_1, J_2). The closed-form expressions for $J_0 = 0$ (19) and $J_0 = 2$ (30) serve as verification checkpoints.

3.4.3 Remark: recovery of the Edland and Diggle formulae

[31] **Setting the run-in count to zero collapses the information matrix to a two-parameter block and yields the slope-comparison variance.**

- With $J_0 = 0$ the δ column vanishes, leaving a 2×2 block on (β, γ).
- Sherman-Morrison cancellations collapse the Woodbury correction against the remaining structure.
- The result is the per-arm variance of the treatment-effect estimator for n participants.

At $J_0 = 0$, the δ column of X is empty and the information matrix reduces to a 2×2 block on (β, γ). Carrying the Woodbury algebra through, the Sherman-Morrison cancellations collapse the Woodbury correction against the $(2,1;1,1)$ structure of $X'R^{-1}X$, and the variance of the treatment effect estimator for n participants per arm becomes

$$\operatorname{Var}(\hat{\gamma})\big|_{J_0=0,J_2=0} = \frac{2}{n}\left[\sigma_b^2 + \frac{\sigma^2}{S_{xx}}\right], \qquad S_{xx} = \sum_{j=0}^{J_1}(t_j - \bar{t})^2, \tag{34}$$

[32] **The boundary case recovers the Edland, Diggle, and Frost formulae, confirming that the general result embeds the longpower routines as strict special cases.**

- The expression is the Ard-Edland slope-comparison formula implemented in the longpower edland routine.
- Setting the slope variance to zero gives compound symmetry and recovers the Diggle formula.
- The Frost formula is the two-visit specialization, so the general result quantifies gains accruing only from run-in or common-close observations.

where the baseline visit at $t_0 = 0$ enters through the change-score transformation. Equation (34) is the slope-comparison formula of[8] implemented in `longpower::edland.linear.power()`. Setting additionally $\sigma_b^2 = 0$ reduces Σ to compound symmetry and recovers the formula of[9] used by `longpower::diggle.linear.power()`. The Frost formula (19) is in turn the $J_1 = 2$ specialization of (34), since $S_{xx} = 2t^2$ at that setting. The general-(J_0, J_1, J_2) result derived in the next subsection therefore embeds both `longpower` closed-form routines as strict boundary cases, with the efficiency gains quantified in Section 5 accruing entirely from the information contributed by $J_0 > 0$ or $J_2 > 0$.

3.5 General formula with common close ($J_2 \geq 0$)

[33] **The final extension adds common-close observations after the treatment period to the change-score vector.**

- J_2 common-close observations follow the treatment period.
- All participants are off-treatment during this phase.
- The total dimension becomes $J = J_0 + J_1 + J_2$.

We now extend to include J_2 common close observations after the treatment period. The total dimension is $J = J_0 + J_1 + J_2$.

During the common close phase (time points $J_1 + 1, \dots, J_1 + J_2$), all participants are off-treatment. The change scores are:

$$c_{i,J_1+s} = [\beta(J_1 + s) + \gamma g_i \cdot J_1 + b_i(J_1 + s)]\, t + (w_{i,J_1+s} - w_{i0}) \quad s = 1, \dots, J_2 \tag{35}$$

[34] **In the common close the treatment effect persists only as an accumulated offset, fixing the design-matrix entry at the treatment-period value.**

- γ enters only through the effect accumulated during the treatment period.
- It does not contribute to the slope during the common close.
- The γ column therefore equals $J_1 t$ for all common-close observations in the treatment group.

The treatment effect γ enters the common close change scores only through the accumulated effect during the treatment period ($\gamma g_i \cdot J_1 \cdot t$), not through the slope during the common close. This means the γ column in the design matrix has the value $J_1 t$ for all common close observations in the treatment group.

The fixed effects design matrix becomes:

$$X_i^{(\text{trt})} = t \begin{pmatrix} -J_0 & 0 & 0 \\ \vdots & \vdots & \vdots \\ -1 & 0 & 0 \\ 0 & 1 & 1 \\ \vdots & \vdots & \vdots \\ 0 & J_1 & J_1 \\ 0 & J_1 + 1 & J_1 \\ \vdots & \vdots & \vdots \\ 0 & J_1 + J_2 & J_1 \end{pmatrix}$$

[35] **The slope and treatment columns of the design matrix encode the post-randomization times and the capped treatment exposure respectively.**

- The β column holds the post-randomization times of the placebo slope across all later visits.
- The γ column rises through the treatment period then stays at J_1 during the common close.
- The γ column is identically zero for placebo participants.

The β column contains the post-randomization time for the placebo slope: $0, \dots, 0, 1, \dots, J_1, J_1{+}1, \dots, J_1{+}J_2$. The γ column is $0, \dots, 0, 1, \dots, J_1, J_1, \dots, J_1$ for treatment participants and all zeros for placebo.

The random effects design remains $Z_i = t \cdot \mathbf{d}$ where $\mathbf{d} = (-J_0, \dots, -1, 1, \dots, J_1, J_1 + 1, \dots, J_1 + J_2)'$.

The residual covariance is still $R = a(\mathbf{I}_J + \mathbf{1}_J\mathbf{1}_J')$ with $J = J_0 + J_1 + J_2$.

[36] **The common-close extension reuses the same Woodbury computation with only three quantities altered.**

- The dimension J increases to include the common-close visits.
- The X matrix changes only in its γ column.
- The elements of the random-effects vector Z are extended accordingly.

The Woodbury computation proceeds identically, with the only changes being the dimension J, the structure of the X matrix (specifically the γ column), and the elements of Z.

We implement the general computation in R below.

4 Power and Sample Size Formulas

For a two-sided test at significance level α with power $1 - \beta_{\text{pow}}$, the required number of participants per group is

$$m = \frac{(z_{\alpha/2} + z_{1-\beta_{\text{pow}}})^2 \cdot \text{Var}_1(\hat{\gamma})}{\Delta^2} \tag{36}$$

where $\text{Var}_1(\hat{\gamma})$ denotes the per-participant-pair variance from the formulas derived above, and Δ is the minimum clinically important treatment effect on the rate of change.

Equivalently, the power for a given sample size is

$$1 - \beta_{\text{pow}} = \Phi\left(\frac{\Delta\sqrt{m}}{\sqrt{\text{Var}_1(\hat{\gamma})}} - z_{\alpha/2}\right) \tag{37}$$

The relative efficiency of a design with (J_0, J_1, J_2) observations compared to a standard design with $(0, J_1, 0)$ is

$$\text{RE}(J_0, J_1, J_2) = \frac{\text{Var}_1(\hat{\gamma})\big|_{0,J_1,0}}{\text{Var}_1(\hat{\gamma})\big|_{J_0,J_1,J_2}} \tag{38}$$

Values greater than 1 indicate that the run-in/common close design requires fewer participants.

[37] **A single run-in observation never worsens precision, but the extra slope parameter it introduces can absorb nearly all of the information it contributes.**

- Adding one run-in observation introduces the parameter δ, enlarging the fixed-effect dimension from 2 to 3.

- Across a wide numerical grid, $\mathrm{Var}(\hat{\gamma})|_{J_0=1}$ never exceeds $\mathrm{Var}(\hat{\gamma})|_{J_0=0}$.
- At some parameter values the gain is negligible, so the benefit of a single run-in observation should be quantified before extending trial duration.

A practical caveat: adding a single run-in observation ($J_0 = 1$) introduces the run-in slope parameter δ into the model, increasing the fixed-effect dimension from 2 to 3. This additional parameter absorbs part of the information contributed by the run-in observation. Numerical evaluation over a wide grid of configurations ($J_1 = 2, \ldots, 6$; σ_b^2/σ^2 from 10^{-4} to 10^3; t from 0.25 to 4) shows that $\mathrm{Var}(\hat{\gamma})|_{J_0=1}$ never exceeds $\mathrm{Var}(\hat{\gamma})|_{J_0=0}$, but the improvement can be negligible: at some parameter values the information from the single run-in observation is almost exactly offset by the cost of estimating δ. A single run-in observation is therefore never harmful to the precision of $\hat{\gamma}$, although the gain it provides should be quantified with the formulas above before committing to a longer trial. This is an instance of the general design principle set out by[17], that the value of an additional measurement occasion in a longitudinal trial depends jointly on the variance components and on the parameters the enlarged model must estimate, so that the number and placement of visits should be chosen by direct evaluation of the resulting precision rather than by a fixed rule.

5 Numerical Examples

5.1 Verification of closed-form results

[38] **The closed-form expressions are validated against the matrix computation in two representative design configurations.**

- The verification confirms agreement between the analytic and matrix-based variances.
- The Frost standard design uses the two-parameter model with two post-baseline visits.
- The two-run-in case exercises the three-parameter model.

We verify that the closed-form expressions agree with the matrix computation. The Frost et al. (2008) result corresponds to the standard design with $J_0 = 0$, $J_1 = 2$ (two post-baseline visits at times t and $2t$), which under the two-parameter model yields $\mathrm{Var}(\hat{\gamma}) = (\sigma^2 + 2\sigma_b^2 t^2)/t^2$. The $J_0 = 2$, $J_1 = 1$ case uses the three-parameter model.

```
a <- 1.0
b <- 0.5
tt <- 1.0

v_frost_matrix <- var_gamma_matrix(0, 2, 0, a, b, tt)
v_frost_closed <- var_gamma_frost(a, b, tt)

v_r2_matrix <- var_gamma_matrix(2, 1, 0, a, b, tt)
```

```r
v_r2_closed <- var_gamma_r2(a, b, tt)

cat('Frost (r=0, k=2) case:\n')
```

```
## Frost (r=0, k=2) case:
```

```r
cat('  Matrix:', round(v_frost_matrix, 6), '\n')
```

```
##   Matrix: 2
```

```r
cat('  Closed:', round(v_frost_closed, 6), '\n')
```

```
##   Closed: 2
```

```r
cat('  Match:',
    all.equal(v_frost_matrix, v_frost_closed), '\n\n')
```

```
##   Match: TRUE
```

```r
cat('r=2, k=1 case:\n')
```

```
## r=2, k=1 case:
```

```r
cat('  Matrix:', round(v_r2_matrix, 6), '\n')
```

```
##   Matrix: 4.666667
```

```r
cat('  Closed:', round(v_r2_closed, 6), '\n')
```

```
##   Closed: 4.666667
```

```r
cat('  Match:',
    all.equal(v_r2_matrix, v_r2_closed), '\n\n')
```

```
##   Match: TRUE
```

```r
cat('Variance comparison across designs:\n')
```

```
## Variance comparison across designs:
```

```r
for (rr in 0:4) {
  v <- var_gamma_matrix(rr, 2, 0, a, b, tt)
  cat(sprintf('  r=%d, k=2: Var = %.4f\n', rr, v))
}
```

```
##   r=0, k=2: Var = 2.0000
##   r=1, k=2: Var = 2.0000
##   r=2, k=2: Var = 1.7910
##   r=3, k=2: Var = 1.5000
##   r=4, k=2: Var = 1.2651
```

5.2 Validation of the general routine

[39] **The Woodbury routine is validated against a direct GLS computation and a seeded Monte Carlo benchmark that both exercise the common-close path.**

- A direct computation assembles and inverts Σ without the Woodbury identity over a grid of designs including $J_2 > 0$.
- Agreement with the Woodbury routine is at the level of numerical round-off.
- A seeded Monte Carlo at $(J_0, J_1, J_2) = (2, 2, 2)$ confirms that the empirical variance of $\hat{\gamma}$ matches the analytic value.

The closed-form checkpoints above both have $J_2 = 0$, so they do not exercise the common-close path. We therefore validate the general routine in two ways. First, we compare the Woodbury computation with a direct GLS evaluation in which $\Sigma = R + \sigma_b^2 ZZ'$ is assembled and inverted without the Woodbury identity, over a grid of designs that includes configurations with $J_2 > 0$. Second, we run a seeded Monte Carlo benchmark at $(J_0, J_1, J_2) = (2, 2, 2)$: change-score vectors are simulated from the generative model, $\hat{\gamma}$ is computed by GLS in each replicate, and the empirical variance is compared with the analytic value.

```r
var_gamma_direct <- function(r, k, f, sigma2, sigma_b2,
                             t_interval) {
  p <- r + k + f
  R <- sigma2 * (diag(p) + matrix(1, p, p))
  des <- build_design(r, k, f, t_interval)
  Sigma <- R + sigma_b2 * des$Z %*% t(des$Z)
  Si <- solve(Sigma)
  info <- t(des$X_trt) %*% Si %*% des$X_trt +
    t(des$X_plc) %*% Si %*% des$X_plc
  solve(info)[des$ncol_X, des$ncol_X]
}

grid <- expand.grid(r = 0:3, k = 1:3, f = 0:3,
                    sigma_b2 = c(0.1, 1, 5),
                    t_int = c(0.5, 1, 2))
grid <- grid[grid$r + grid$k + grid$f >= 3, ]
diffs <- mapply(
  function(r, k, f, sb2, tt) {
    abs(var_gamma_matrix(r, k, f, 1.0, sb2, tt) -
          var_gamma_direct(r, k, f, 1.0, sb2, tt))
  },
  grid$r, grid$k, grid$f, grid$sigma_b2, grid$t_int
)
cat('Direct GLS vs Woodbury routine over', nrow(grid),
    'configurations (including f > 0):\n')
```

```
## Direct GLS vs Woodbury routine over 396 configurations (including f > 0):
```

```r
cat('  Maximum absolute difference:',
    format(max(diffs), digits = 3), '\n\n')
```

```
737 ##   Maximum absolute difference: 3.38e-14
```

```r
set.seed(20260418)
r_v <- 2; k_v <- 2; f_v <- 2
s2_v <- 1; sb2_v <- 1; t_v <- 1
m_v <- 200; n_sim <- 2000
des <- build_design(r_v, k_v, f_v, t_v)
p_v <- r_v + k_v + f_v
Sigma <- s2_v * (diag(p_v) + matrix(1, p_v, p_v)) +
  sb2_v * des$Z %*% t(des$Z)
Si <- solve(Sigma)
cR <- chol(Sigma)
info <- m_v * (t(des$X_trt) %*% Si %*% des$X_trt +
                 t(des$X_plc) %*% Si %*% des$X_plc)
theta <- c(0.2, -0.5, 0.3)
gamma_hat <- numeric(n_sim)
for (s in seq_len(n_sim)) {
  xty <- 0
  for (X in list(des$X_trt, des$X_plc)) {
    mu <- as.numeric(X %*% theta)
    Y <- matrix(rep(mu, m_v), m_v, p_v, byrow = TRUE) +
      matrix(rnorm(m_v * p_v), m_v, p_v) %*% cR
    xty <- xty + t(X) %*% Si %*% colSums(Y)
  }
  gamma_hat[s] <- solve(info, xty)[3]
}
cat('Monte Carlo at (J0, J1, J2) = (2, 2, 2), m =', m_v,
    'per group,', n_sim, 'replicates:\n')
```

```
738 ## Monte Carlo at (J0, J1, J2) = (2, 2, 2), m = 200 per group, 2000 replicates:
```

```r
cat('  Empirical   m * Var(gamma_hat):',
    round(m_v * var(gamma_hat), 4), '\n')
```

```
739 ##   Empirical   m * Var(gamma_hat): 1.1039
```

```r
cat('  Theoretical (Woodbury):        ',
    round(var_gamma_matrix(r_v, k_v, f_v, s2_v, sb2_v,
                           t_v), 4), '\n')
```

```
740 ##   Theoretical (Woodbury):        1.1344
```

Table 1: Required sample size per group ($J_1 = 2$, $J_2 = 0$, $\Delta = 0.5$, 90% power, two-sided $\alpha = 0.05$) as a function of the number of run-in observations J_0, for three levels of between-subject slope variability.

J_0	Low	Medium	High
0	51	127	463
1	44	119	214
2	44	91	113
3	43	70	76
4	42	56	59
5	40	48	49
6	38	42	43

Table 2: Required sample size per group ($J_0 = 2$, $J_1 = 2$, $\Delta = 0.5$, 90% power) as a function of common close observations J_2.

J_2	Low	Medium	High
0	44	91	113
1	34	69	80
2	29	48	52
3	25	35	37
4	22	27	28

5.3 Sample size as a function of run-in observations

5.4 Sample size as a function of common close observations

5.5 Relative efficiency surface

5.6 Application: Alzheimer's disease imaging trial

[40] **The Alzheimer's application uses MIRIAD-derived parameters with the correctly combined within-visit residual, which is essential to reproduce Frost's reported sample sizes.**

- The slope variance and combined within-visit residual are taken from Frost's MIRIAD estimates.
- The example assumes a one-year observation interval and a treatment effect of one quarter.
- Using only the small measurement-error term understates the residual by two orders of magnitude and yields degenerate sample sizes.

Using variance parameters estimated from the MIRIAD dataset[2]: $\sigma_b^2 = 0.9745$ *(between-subject slope variance) and a combined within-visit residual* $\sigma^2 = \sigma_u^2 + \sigma_\varepsilon^2 = 0.3338 + 0.0025 = 0.3363$*, formed per Eq. (2) from the within-subject between-visit component (*$\sigma_u^2 = 0.3338$*) and the additional between-scan measurement error (*$\sigma_\varepsilon^2 = 0.0025$*) of Frost's Table II. We take observation interval* $t = 1$ *year and treatment effect* $\Delta = 0.25$ *(25% reduction in atrophy rate relative to placebo rate of approximately 1.0% brain*

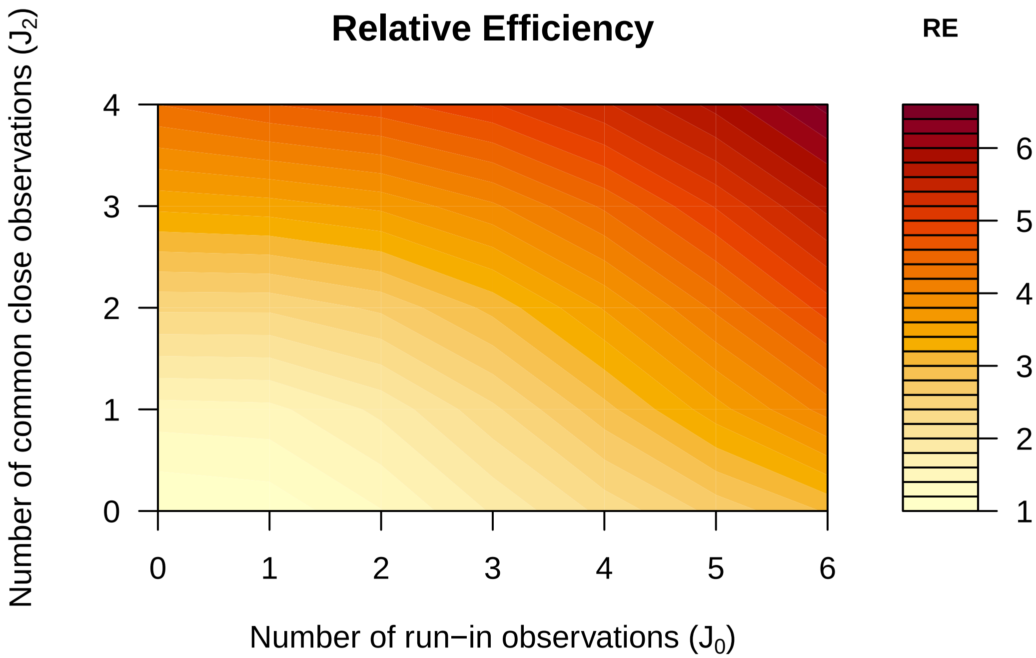

Figure 1: (#fig:fig_efficiency)Relative efficiency of run-in and common close designs compared to the standard design ($J_0 = 0$, $J_2 = 0$) with the same number of post-randomization observations ($J_1 = 2$). Parameters: $\sigma^2 = 1$, $\sigma_b^2 = 1$, $t = 1$.

volume loss per year). Using the combined residual, the conventional design reproduces Frost's reported sample sizes (Table III); using only the 0.0025 *measurement-error term, as in earlier drafts, understates the residual by two orders of magnitude and yields degenerate sample sizes.*

```
## AD imaging trial sample sizes (per group):
##   Standard (r=0, k=2, f=0): 385
##   Run-in   (r=2, k=2, f=0): 145  (62% reduction)
##   Full     (r=2, k=2, f=2): 69  (82% reduction)
```

6 Discussion

[41] **The work formalizes and extends Frost's framework with explicit variance formulas for run-in and common-close designs.**

- The results provide explicit treatment-effect variance formulas.
- They cover designs incorporating both run-in and common-close observations.
- Three substantive points warrant further discussion.

The results presented here formalize and extend the framework of[2] *by providing explicit variance formulas for the treatment effect estimator under designs that incorporate run-in and common close observations. Three points merit discussion.*

Table 3: Sample size per group for AD imaging trial ($J_1 = 2$ post-randomization visits, $\Delta = 0.25$, 90% power). All $J_1 = 2$ configurations ordered by sample size; the $(0, 2, 0)$ row is the standard-design baseline.

J_0	J_1	J_2	m per group	Total visits
4	2	2	49	9
3	2	2	58	8
4	2	1	60	8
2	2	2	69	7
3	2	1	77	7
0	2	2	77	5
1	2	2	77	6
4	2	0	78	7
3	2	0	100	6
2	2	1	104	6
1	2	1	138	5
2	2	0	145	5
0	2	1	150	4
1	2	0	247	4
0	2	0	385	3

[42] **The benefit of run-in observations is governed by the slope-to-noise ratio and is largest for outcomes such as neuroimaging.**

- The efficiency gain depends on the ratio of slope variance to residual variance.
- When that ratio is large, even one or two run-in observations cut sample sizes substantially.
- For cognitive outcomes with a less favorable ratio, gains are modest and must be weighed against longer study duration.

When run-in observations are most beneficial. *The efficiency gain from run-in observations depends on the ratio $\sigma_b^2 t^2/\sigma^2$. When between-subject slope variability is large relative to residual variance (as in neuroimaging outcomes), even one or two run-in observations yield substantial sample size reductions. For cognitive outcomes, where this ratio is typically less favorable, the gains are more modest and the cost of extended study duration must be weighed against the sample size reduction.*

[43] **Common-close observations sharpen the treatment-effect estimator at low cost, with diminishing returns beyond the first few visits.**

- They inform individual trajectories without keeping participants on treatment.
- Both groups contribute to slope-variability estimation once treatment has ceased.

- Marginal benefit diminishes, but the first one or two visits give meaningful reductions when the slope-to-noise ratio is large.

Common close as a low-cost efficiency gain. *The common close phase provides information about individual trajectories without requiring participants to remain on treatment. Since treatment has ceased, both groups contribute to estimation of the underlying slope variability, effectively tightening the precision of the treatment effect estimator. The marginal benefit of each additional common close observation diminishes (as with run-in observations), but the first one or two common close visits typically provide meaningful sample size reductions, particularly when* $\sigma_b^2 t^2/\sigma^2$ *is large.*

[44] **The derived formulas are consistent with Wang's two-period model while making the three phases explicit and separately optimizable.**

- Wang's common-slope scenario corresponds to setting the run-in and placebo slopes equal.
- Their different-slope scenario corresponds to the general case treated here.
- The present formulation separates run-in, treatment, and common-close phases, enabling per-phase optimization.

Relationship to the two-period model. *The formulas derived here are consistent with the two-period linear mixed effects model of*[3]*. Their Scenario 1 (common slope across periods) corresponds to setting* $\delta = \beta$ *in our model, while their Scenario 2 (different slopes) corresponds to the general case. The advantage of our formulation is the explicit separation of run-in, treatment, and common close phases, enabling direct optimization of the number of observations in each phase.*

[45] **The derivations rest on simplifying assumptions whose relaxation is feasible but typically forfeits closed-form simplicity.**

- They assume equally spaced observations, linear trajectories, and a random intercept-plus-slope structure.
- Extensions to unequal spacing, nonlinear progression, or richer random effects remain within the Woodbury framework but lose simple closed forms.
- No dropout is assumed, though the Dawson-Lagakos pattern-mixture approach can adjust for anticipated attrition.

Limitations. *The derivations assume equally spaced observations, linear trajectories, and a random intercept plus random slope covariance structure. Extensions to unequal spacing, nonlinear progression, and more complex random effects structures (e.g., random intercept-slope correlation) are possible within the same Woodbury identity framework but yield more complex expressions that may not admit simple closed forms. The formulas also assume no dropout; in practice, the Dawson-Lagakos pattern-mixture approach*[2] *can be applied to adjust for anticipated attrition.*

7 Conflict of interest

The author declares no conflicts of interest.

8 Funding

This work received no specific funding.

9 Data availability statement

[46] **All materials supporting the paper are openly available in the project repository.**

- These include the R package, the four companion manuscripts, and the Mathematica derivation scripts.
- The shared bibliography is likewise available.
- Specific paths are enumerated in the Reproducibility section.

The `runinpower` R package, the four companion manuscripts of this compendium, the Mathematica symbolic-derivation scripts, and the shared bibliography are openly available in the project repository. Specific paths are listed in the Reproducibility section below.

10 Reproducibility

[47] **The manuscript build relies on the in-package source together with a small set of testing and rendering packages.**

- Rendering used R version 4.4.x.
- The principal dependencies are the in-package source plus tinytest, rmarkdown, and bookdown.
- Supporting Mathematica symbolic derivations reside in the analysis scripts directory.

This manuscript was rendered with R 4.4.x. Principal packages used are the in-package `runinpower` source (nine source files: allocation, ancova, averaged, covariance, covariance-ar1, design, power, replicated, variance), `tinytest` (testing harness), and `rmarkdown` plus `bookdown` for the manuscript build. Mathematica symbolic derivations supporting the Woodbury-based variance formulas are at `analysis/scripts/`.

[48] **The variance results are deterministic, and the Monte Carlo benchmark reported in the validation subsection uses a fixed seed.**

- Given fixed input parameters the formulas exercise no random-number generation.

- The validation subsection compares the Woodbury routine with a direct GLS computation over a grid that includes common-close configurations.
- Its Monte Carlo benchmark at $(J_0, J_1, J_2) = (2, 2, 2)$ sets the seed 20260418 at its entry point.

Random-number generator. *The variance formulas are deterministic given fixed input parameters; no RNG is exercised in their evaluation. The validation chunk in the Numerical Examples section first compares the Woodbury routine with a direct GLS computation, in which Σ is assembled and inverted without the Woodbury identity, over a grid of designs that includes common-close configurations ($J_2 > 0$). It then runs a Monte Carlo benchmark at $(J_0, J_1, J_2) = (2, 2, 2)$ with 2000 replicates of $m = 200$ participants per group, setting `set.seed(20260418)` at its entry point and comparing the empirical variance of $\hat{\gamma}$ with the analytic value.*

[49] **The repository organizes the package source, derivation scripts, companion manuscripts, and bibliography into well-defined locations.**

- The R package source and the Mathematica scripts each have a dedicated directory.
- The three companion manuscripts share a report subtree.
- This manuscript source and the shared bibliography sit in the run-in power report directory.

Data and code availability. *The R package source is at `R/`. Mathematica scripts are at `analysis/scripts/`. The companion manuscripts are at `analysis/report/{02,03,04}-*/`. The manuscript source and shared bibliography (`../references.bib`) are at `analysis/report/01-runin-power/`.*

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928 *Rendered on 2026-08-08 at 18:05 PDT.*

929 *Source:* `/Users/zenn/Library/CloudStorage/Dropbox/prj/res/04-runin-power-analysis/runinpower/a`